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School of Computer Science and Engineering

Department of Computer Science and Engineering

AI Teaching Assistant Using the Socratic Method for Conceptual Learning

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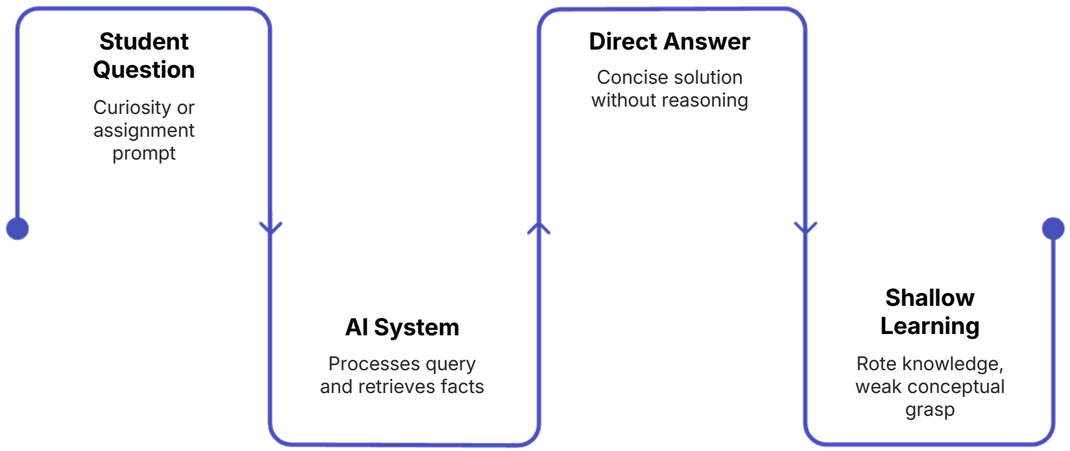
Introduction

Problem Context

- Rise of AI in education
- Direct answers lead to rote learning
- Weak conceptual clarity
- Lack of critical thinking

Current Learning Flow



This linear flow often leads to superficial understanding.

Why Socratic Method?

Key Principles

- Socratic method promotes deep thinking
- Learning through structured questioning
- Encourages reasoning and self-explanation
- Difficult to implement at scale in classrooms

AI Approaches Compared

Traditional AI

Question → Answer

Socratic AI

Question → Guided Questions → Student Reasoning → Understanding

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Example: Understanding Recursion

Traditional AI Approach

Question: "What is recursion?"

Answer: "Recursion is a function that calls itself with a base case and recursive case."

Socratic AI Approach

Question: "What is recursion?"

Guided Questions:

1. "Can you think of a real-world example where something repeats itself?"
2. "How would you solve this problem if you broke it into smaller versions?"
3. "What would stop the repetition from going on forever?"

Student realizes: "Oh! Recursion is breaking a problem into smaller versions of itself!"

Literature Review

Research Papers Reviewed

Paper	Approach	Limitation
Automated Socratic Tutoring (2021)	Rule-based question templates	Not LLM-based
Conversational AI Tutors (2023)	Inquiry-driven	Shallow adaptation
Bloom’s Taxonomy AI (2020)	Level-based questioning	No LLM integration

Literature Review

Identified Gaps

- Most systems provide answers, not guided reasoning.
- Limited adaptive Socratic questioning.
- Lack of integration: LLM + Bloom’s + Dialogue control.
- No structured framework combining all three.

Research Gap:

Existing AI tutors lack a fully adaptive Socratic questioning framework powered by LLMs for deep conceptual learning.

Problem Statement

The Problem

Students depend on direct-answer AI tools, leading to shallow understanding and poor conceptual clarity.

Solution Needed

There is a need for an adaptive AI teaching assistant that promotes deep reasoning using structured Socratic questioning.

Objectives

Design an AI-based Socratic tutoring framework

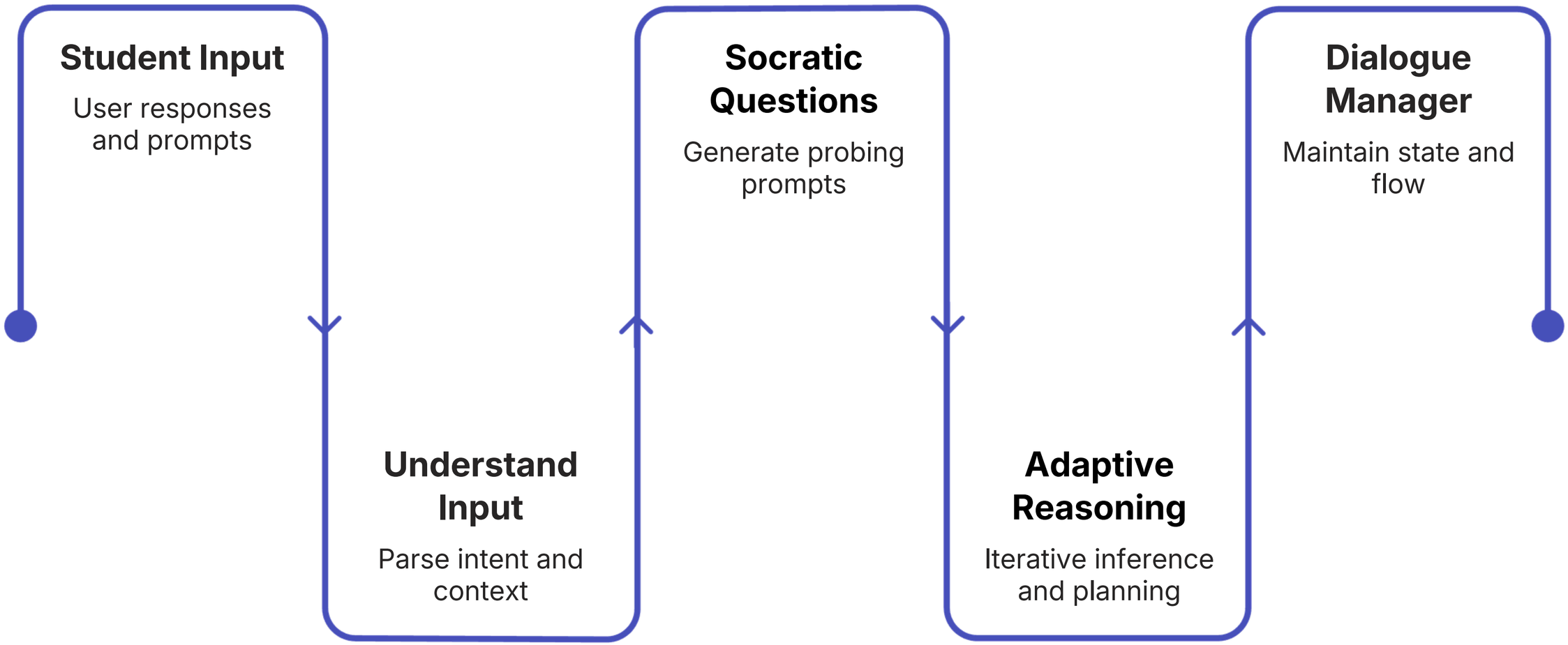
Integrate LLM with Bloom's taxonomy-based adaptation

Develop prototype architecture and interaction flow

Evaluate conceptual learning effectiveness

Proposed Methodology

System Architecture



Proposed Methodology

Question Design Strategy

Clarification Questions

"What do you mean by...?"

Why-Questions

"Why does this work?"

Counterexample Prompts

"What if...?"

Application Prompts

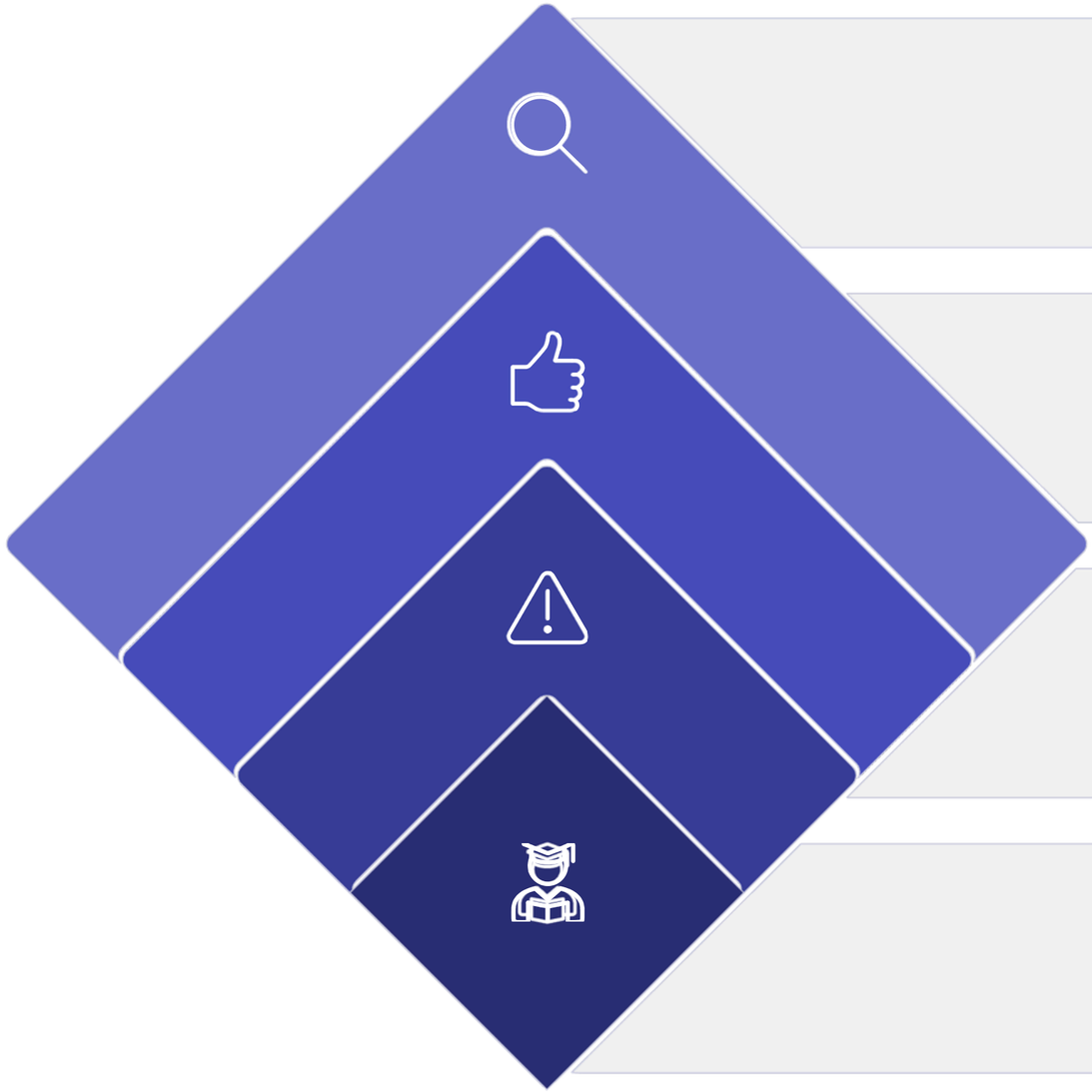
"How would you apply...?"

Reflective Questions

"What did you learn?"

Proposed Methodology

Adaptive Difficulty Logic



Misconception Detected

Incorrect pattern found

Confident

Quick correct answers

Struggles

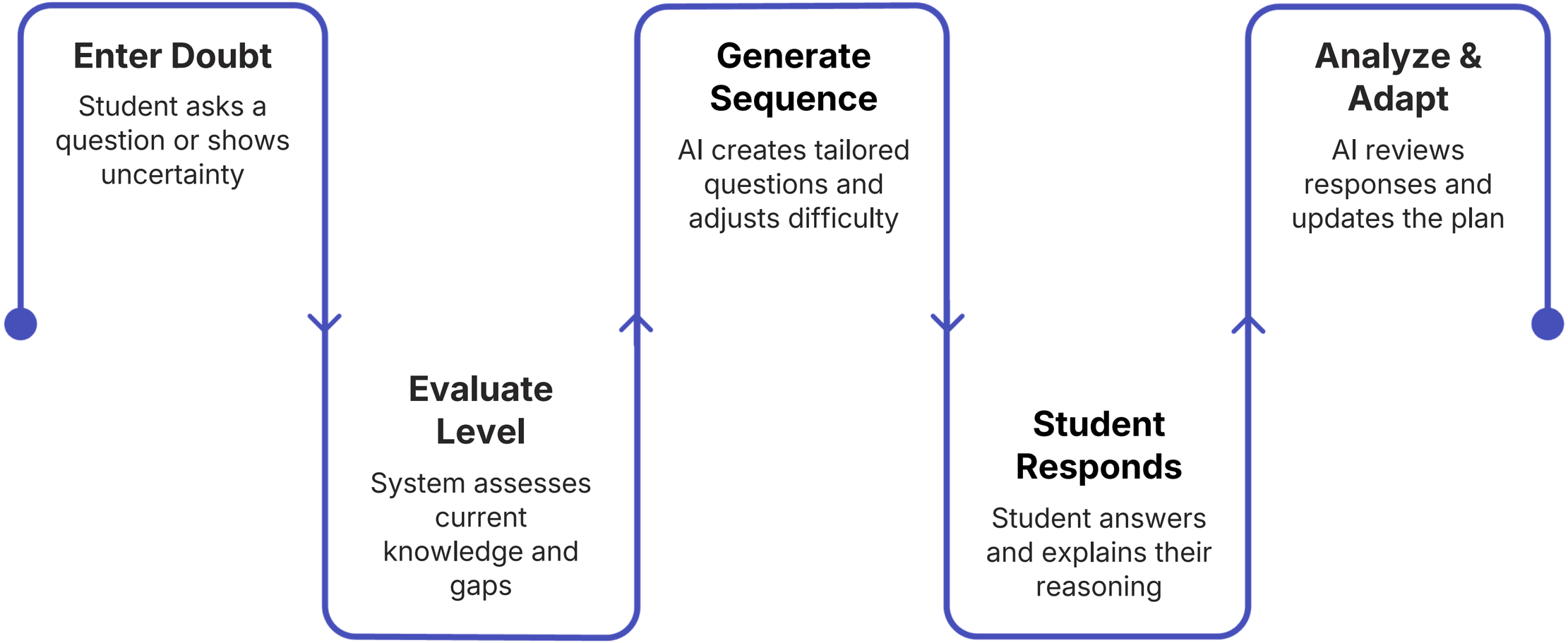
Low accuracy or hesitation

Student Response

Capture interaction outcome

Our adaptive difficulty logic dynamically adjusts the learning path based on student interaction.

Interaction Flow



Results & Progress Summary

Research Gap Identification	Gap between direct-answer AI and adaptive Socratic AI	Completed
Literature Survey	Comparative study of existing tutoring systems	Completed
System Architecture	Modular LLM-based Socratic tutor design	Completed
Question Framework	Structured Socratic question types defined	Completed
Adaptive Logic Model	Bloom's taxonomy-based difficulty progression	Designed
Prototype Implementation	Not yet developed	Planned

Outcomes

Current Semester Achievements

- Identified research gap in AI tutoring systems
- Designed a structured Socratic questioning framework
- Developed system architecture for LLM-based adaptive tutor
- Defined Bloom's taxonomy–based difficulty progression
- Prepared implementation roadmap for prototype development

Planned Outcomes

- Development of functional prototype
- Integration of LLM-based question generation
- Testing with sample student scenarios
- Drafting research paper for conference submission

References

Research Papers

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2. B. Author et al., "Conversational AI Tutors for Inquiry-Based Learning," 2023.
3. C. Author et al., "Guided Questioning with Large Language Models for Deeper Learning," 2024.

Foundational Frameworks

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AI Learning Systems & Frameworks

1. Harvard University, "CS50 Rubber Duck AI Debugging Assistant," 2023.
2. Khan Academy, "Khanmigo – AI Learning Assistant," 2023.
3. Google Research, "LearnLM: Learning-Optimized Large Language Model," 2024.
4. OpenAI, "GPT-4o Education Framework," 2024.

Thank You